ODE Is All You Need

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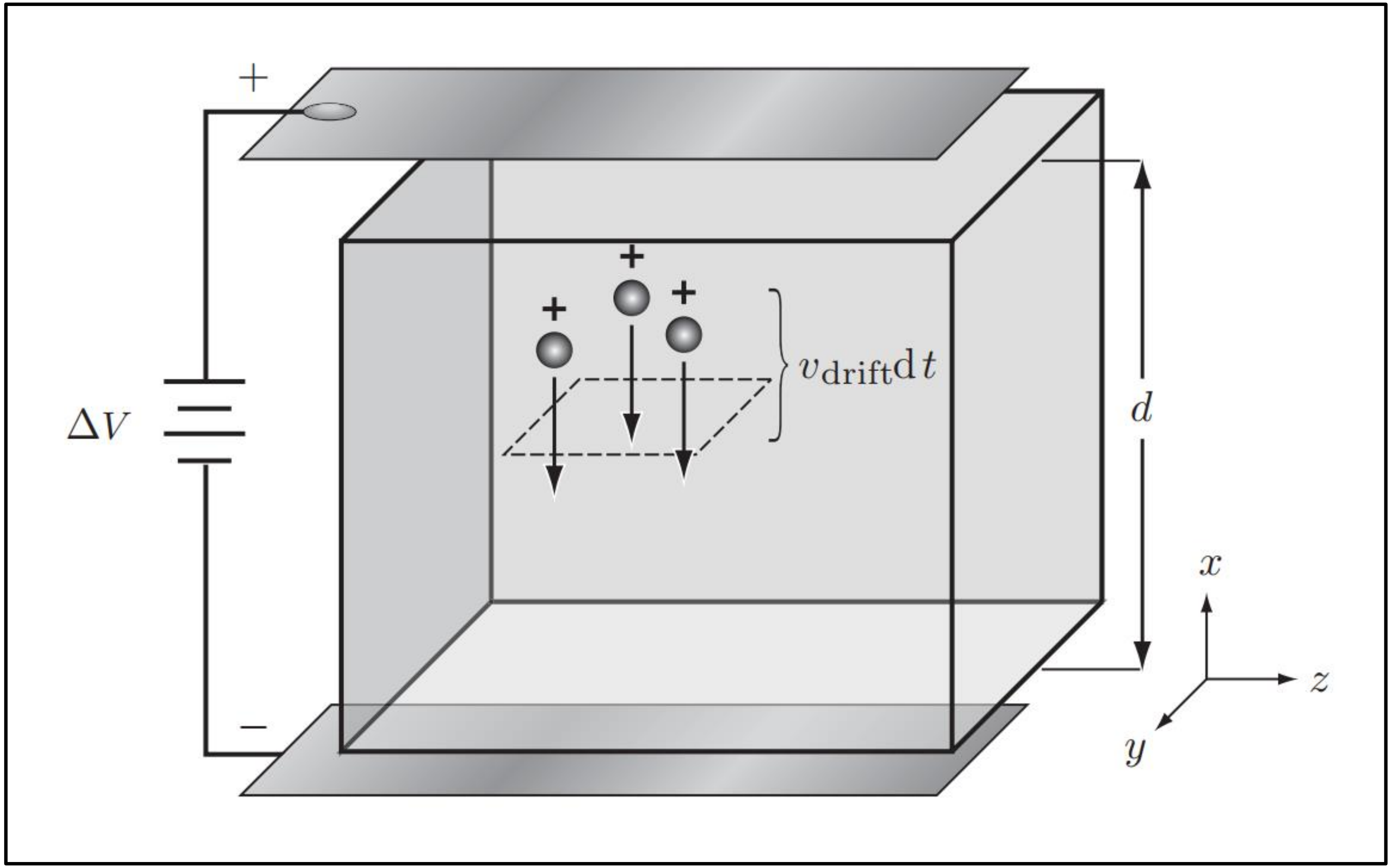
Outline

- Nernst equation
- Lapicque model/Integrate and fire model
- Hodgkin-Huxley model
- others

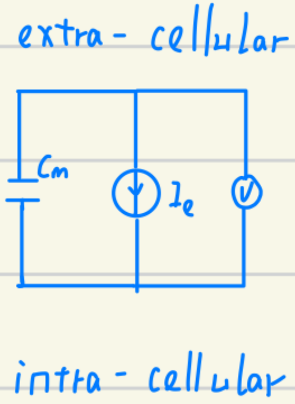
Nernst



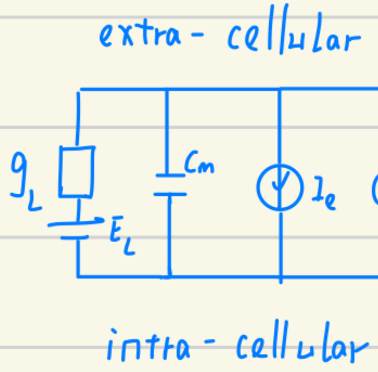
Integrate and fire model

- non-leaky
- leaky

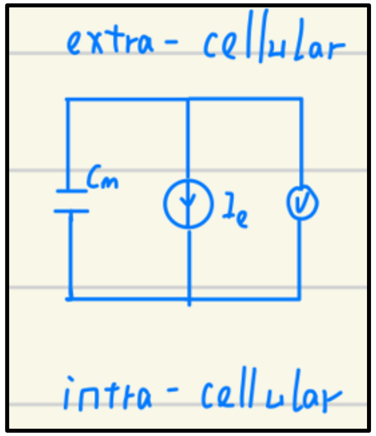
- non-leaky
  - $$\begin{cases} C\dfrac{dV}{dt} = I(t) \\ V(t^-_{spike}) = V_{threshold} \\ V(t^+_{spike}) = V_{resting} \end{cases}$$

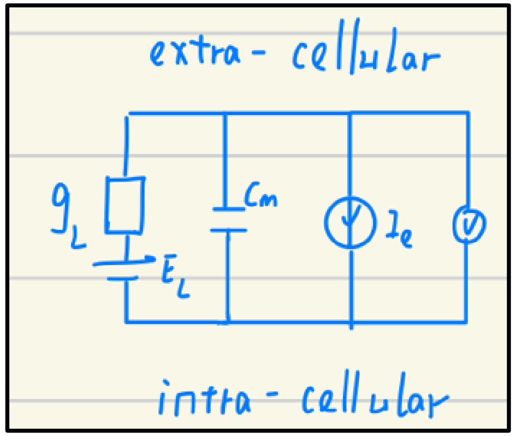


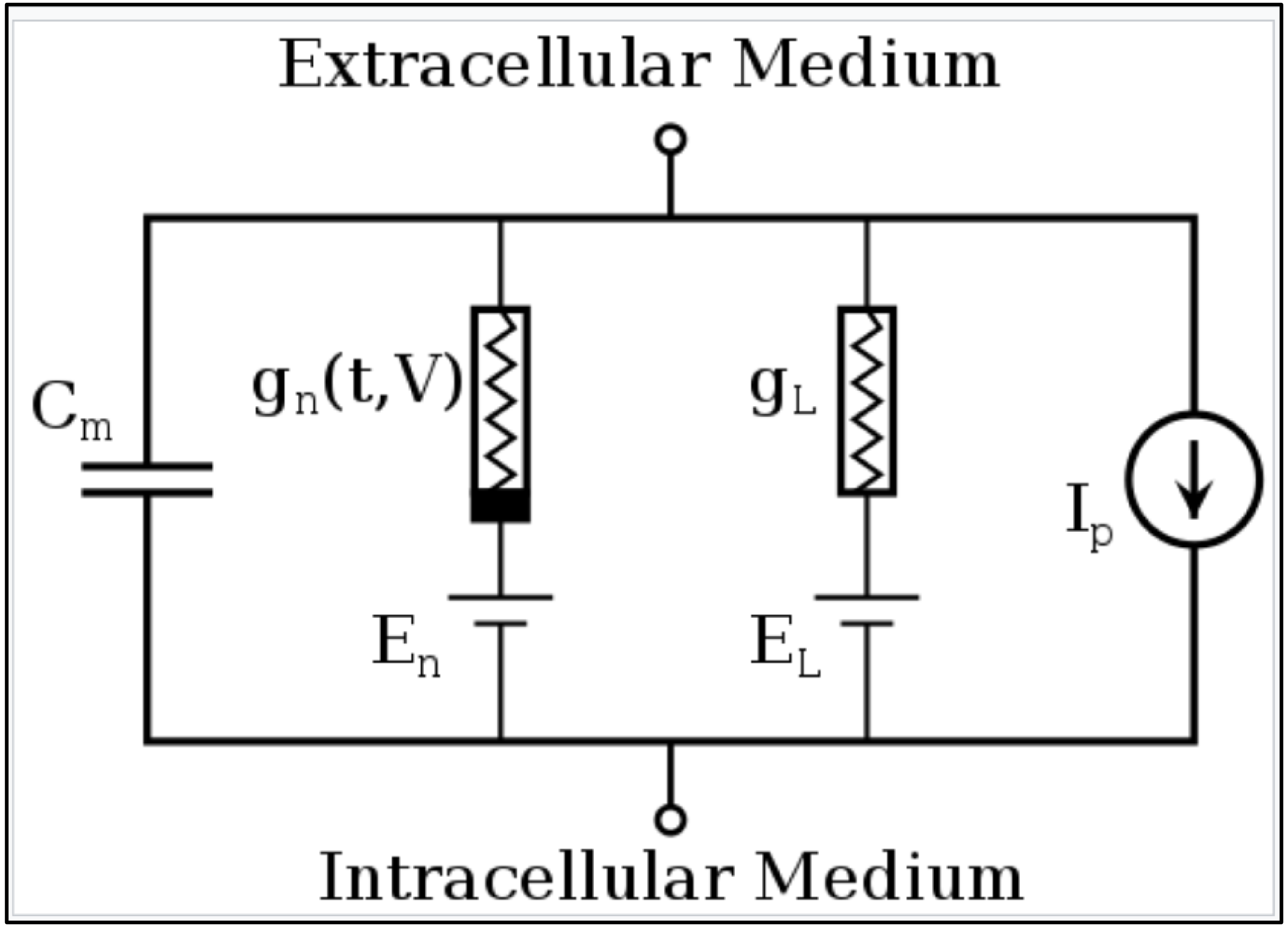
- leaky
  - $$\begin{cases} C\dfrac{dV}{dt} = I(t) + (-g_L(V-E_L)) \\ V(t^-_{spike}) = V_{threshold} \\ V(t^+_{spike}) = V_{resting} \end{cases}$$



Hodgkin-Huxley model







HH: different types of neuron firing

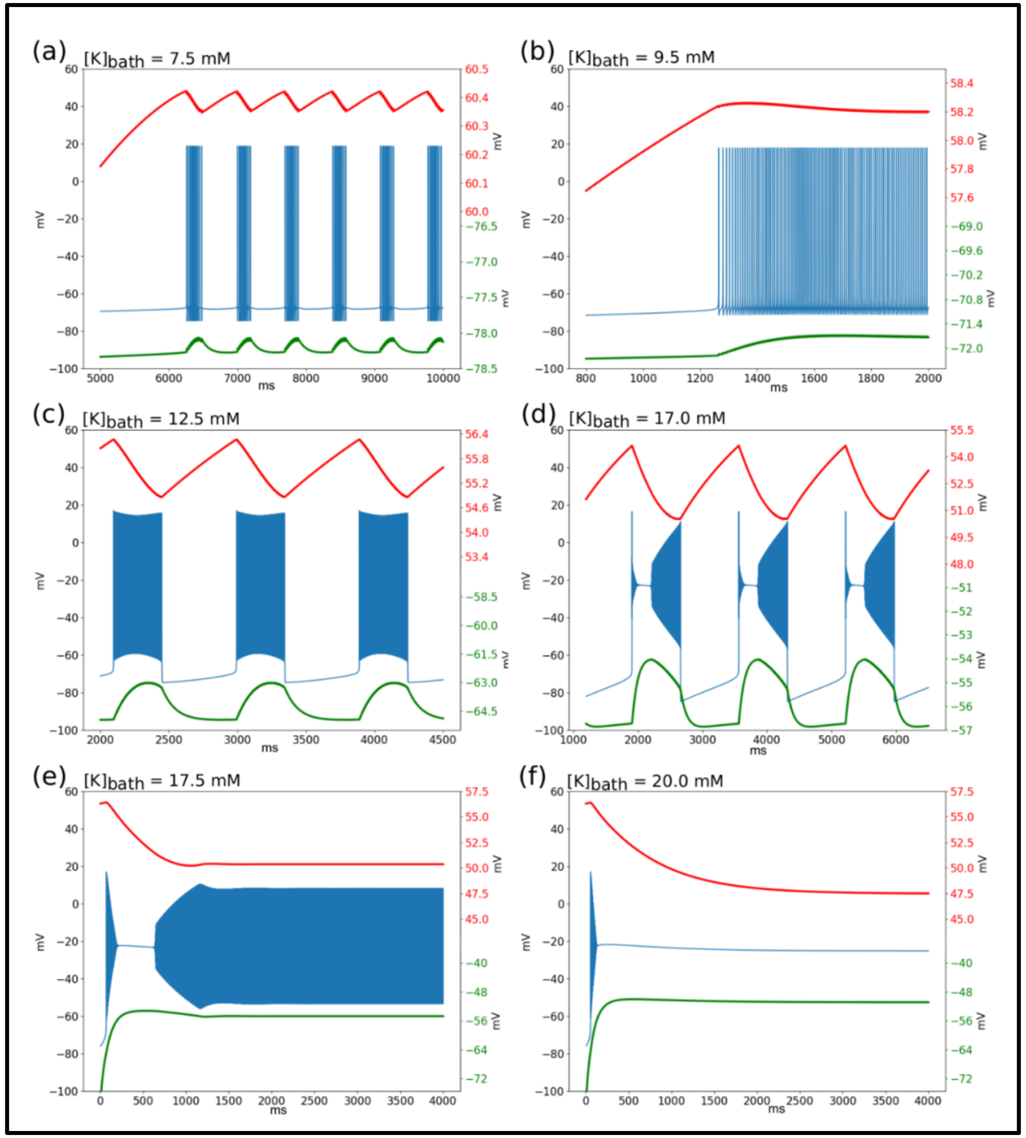
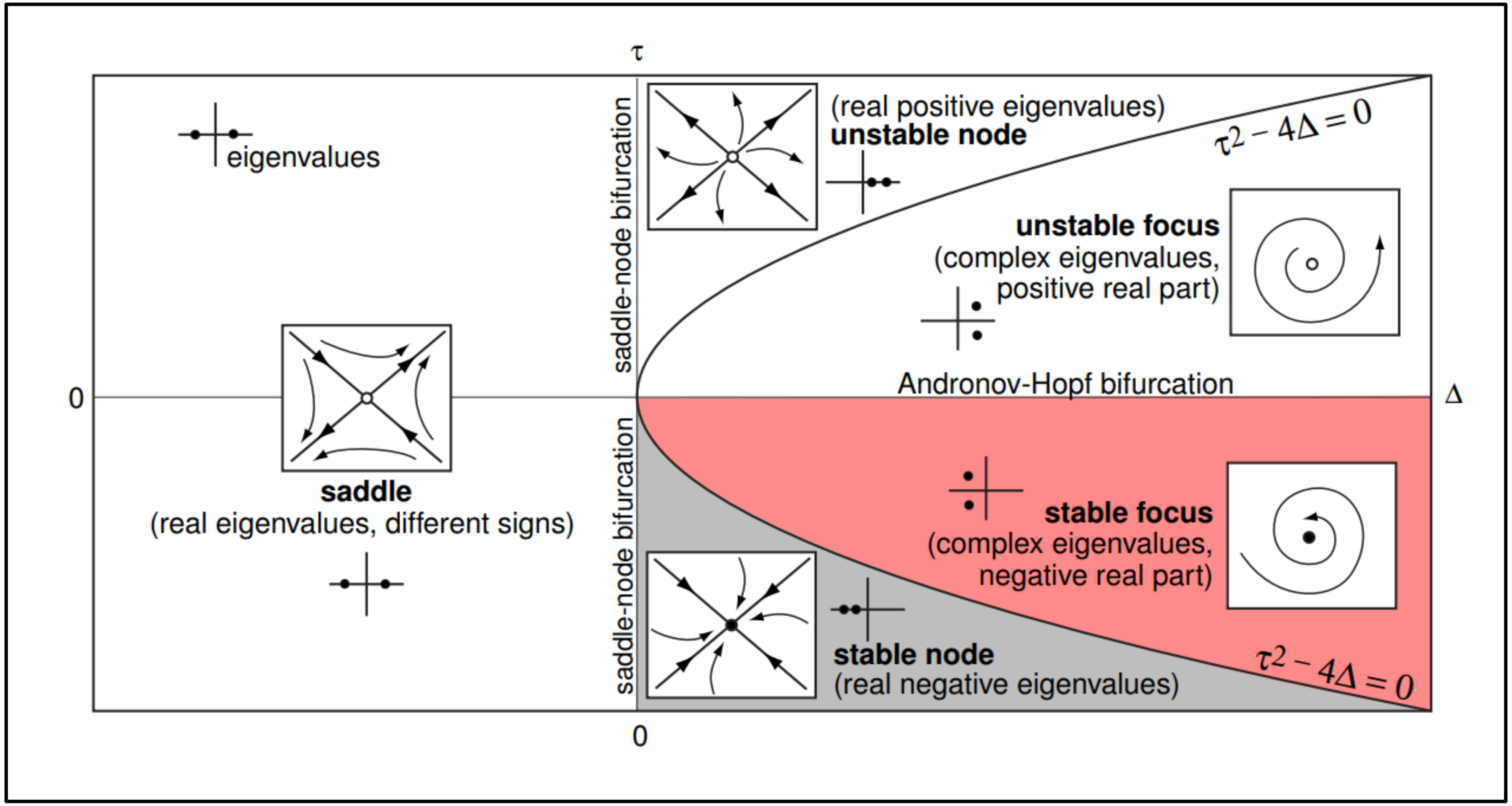


Figure 2: qualitative mode of behavior of the membrane potential and Nernst potentials. In blue: time series of the membrane potential V_m for the following patterns of activity: (a) Spike train, (b) Tonic spiking (TS), (c) Bursting, (d) Seizure-like event (SLE), (e) sustained ictal activity, (f) Depolarization Block (DB). In red: Nernst potential of sodium, in green: Nernst potential of potassium with specific Y axis on the right side. If the value of $[K]_{bath}$ stays below 6 mM, the system remains in resting state around -72 mV. Specific patterns of activities start to appear with a diminution of the Nernst potential of sodium and an increase of the Nernst potential of potassium. When periodic events are occurring (panels c and d), oscillations can also be observed in the Nernst potential of both ions.

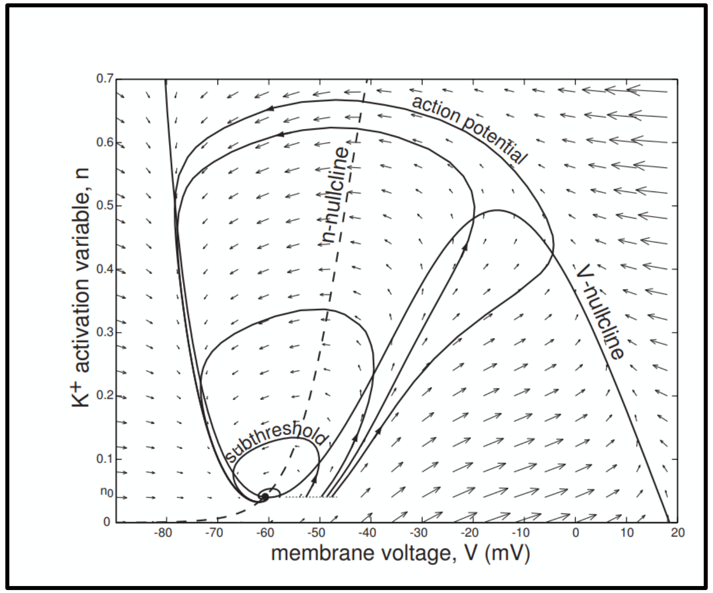
cite: A unified physiological framework of transitions between seizures sustained ictal activity and depolarization block at the single neuron level

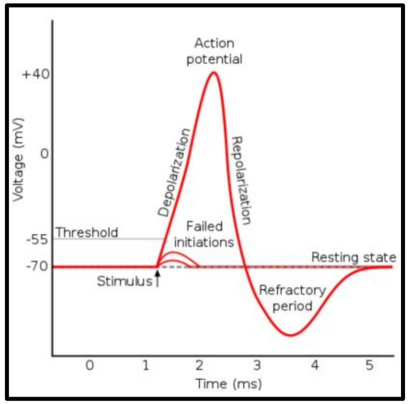
HH: 4ODEs->2ODES



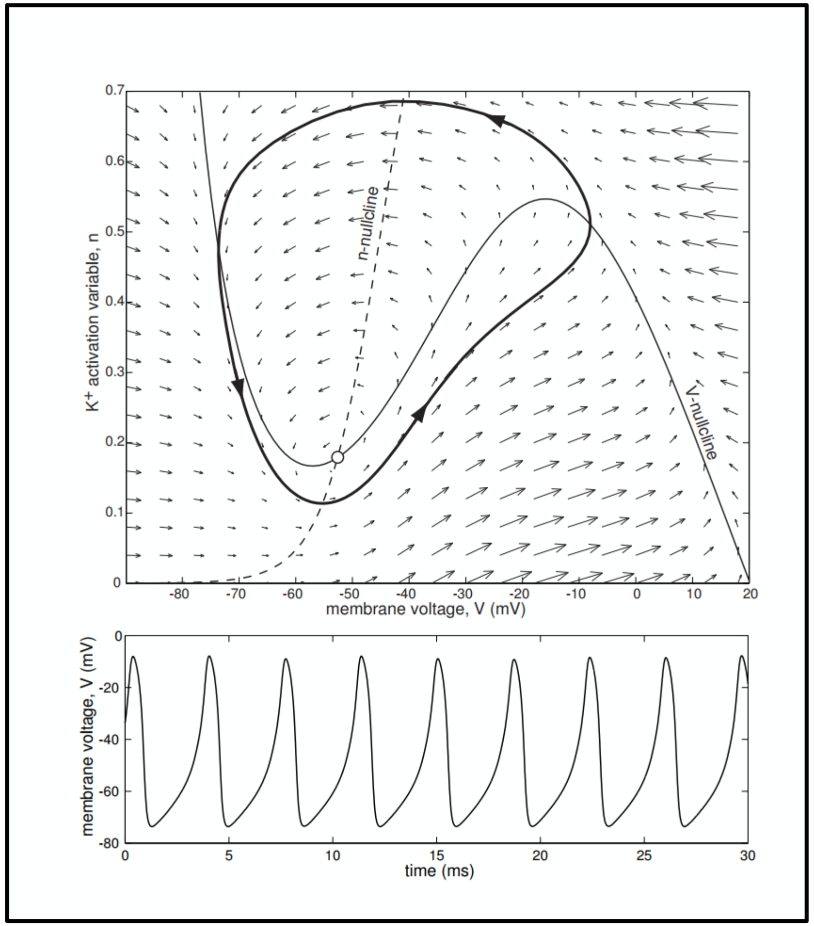
Nullcline and Vector Field

I < threshold





I > threshold



Why HH only uses Na+ and K+?

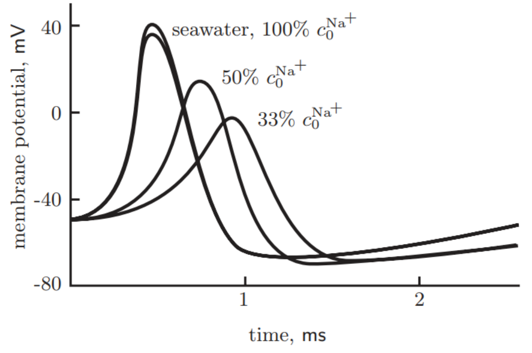


Figure 12.7: (Experimental data.) The role of sodium in the conduction of an action potential. One of the top traces was taken on a squid axon in normal seawater before exposure to low sodium. In the middle trace, external sodium was reduced to one-half that in seawater, and in the bottom trace, to one third. (The other top trace was taken after normal seawater was restored to the exterior bath.) The data show that the peak of the action potential tracks the sodium Nernst potential across the membrane, supporting the idea that the action potential is a sudden increase in the axon membrane's sodium conductance. [After Hodgkin & Katz, 1949]

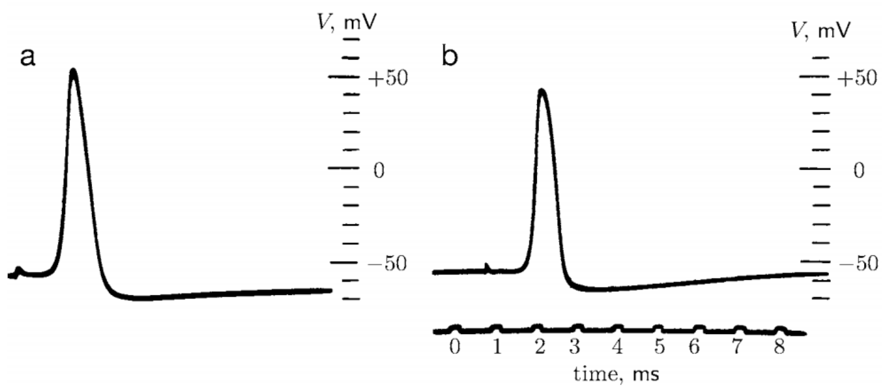


Figure 12.13: (Oscilloscope traces.) Perhaps the most remarkable experiment described in this book. (a) Action potential recorded with an internal electrode from an axon whose internal contents have been replaced by potassium sulfate solution. (b) Action potential of an intact axon, with same amplification and time scale. [From Baker et al., 1962.]

Is it ok to use circuit as a model of membrane?

- Well…